

Leishmaniasis

- Chronic inflammatory disease of
 - skin
 - mucous membrane
 - viscera

- vector-borne zoonosis
- common **reservoir** hosts → rodents and canids
- humans are **incidental hosts**

- Cause
 - *Leishmania* parasite
 - Obligate intracellular protozoan parasite
 - Multiplies in macrophages
 - Infection exacerbated by any condition interfering with T cell function

Epidemiology

- 20 million people affected worldwide
- **Endemic**: - Middle east, South Asia, Africa and Latin America
- May cause **epidemic, visceral disease**, resulting in death



Transmission

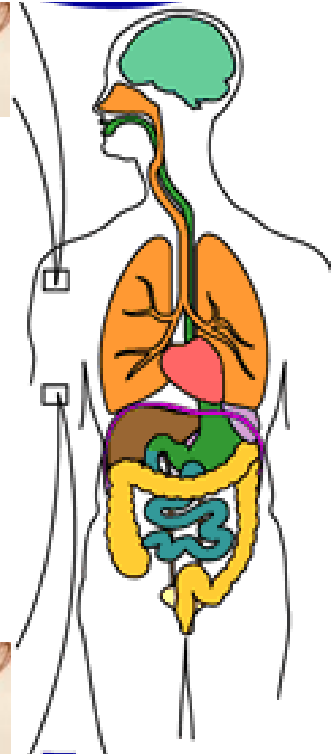
- Vector: - **infected Sandfly**
 - *Phlebotomus* spp → old world
 - *Lutzomia* spp → new world

Life Cycle

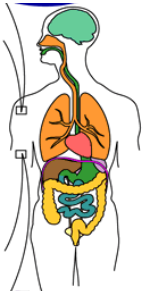
1- Sandfly bites animal and ingests blood infected with *Leishmania*



2- Sandfly bites human and injects *Leishmania* into skin



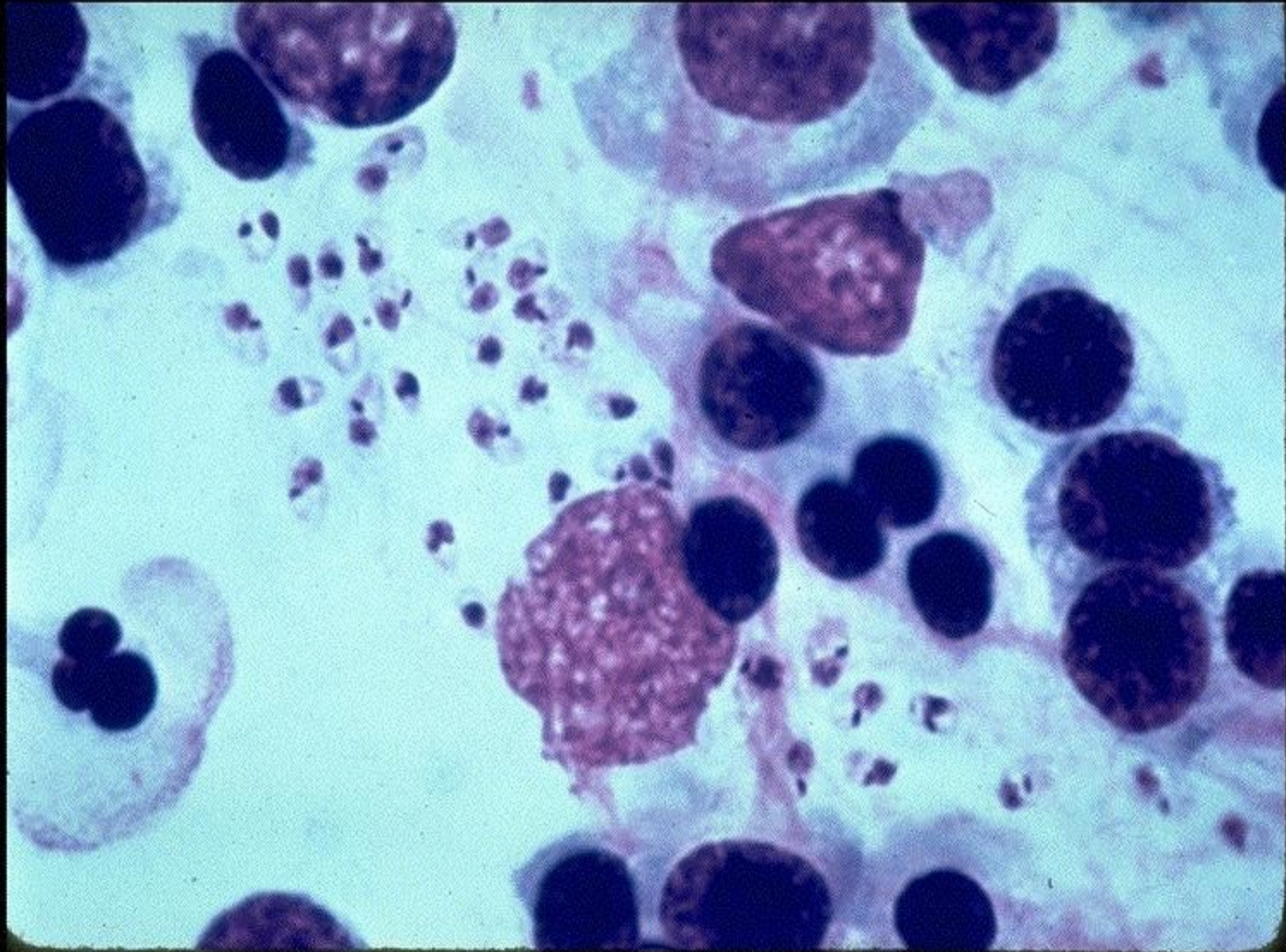
4- Cycle continues when sandfly bites another human or animal reservoir

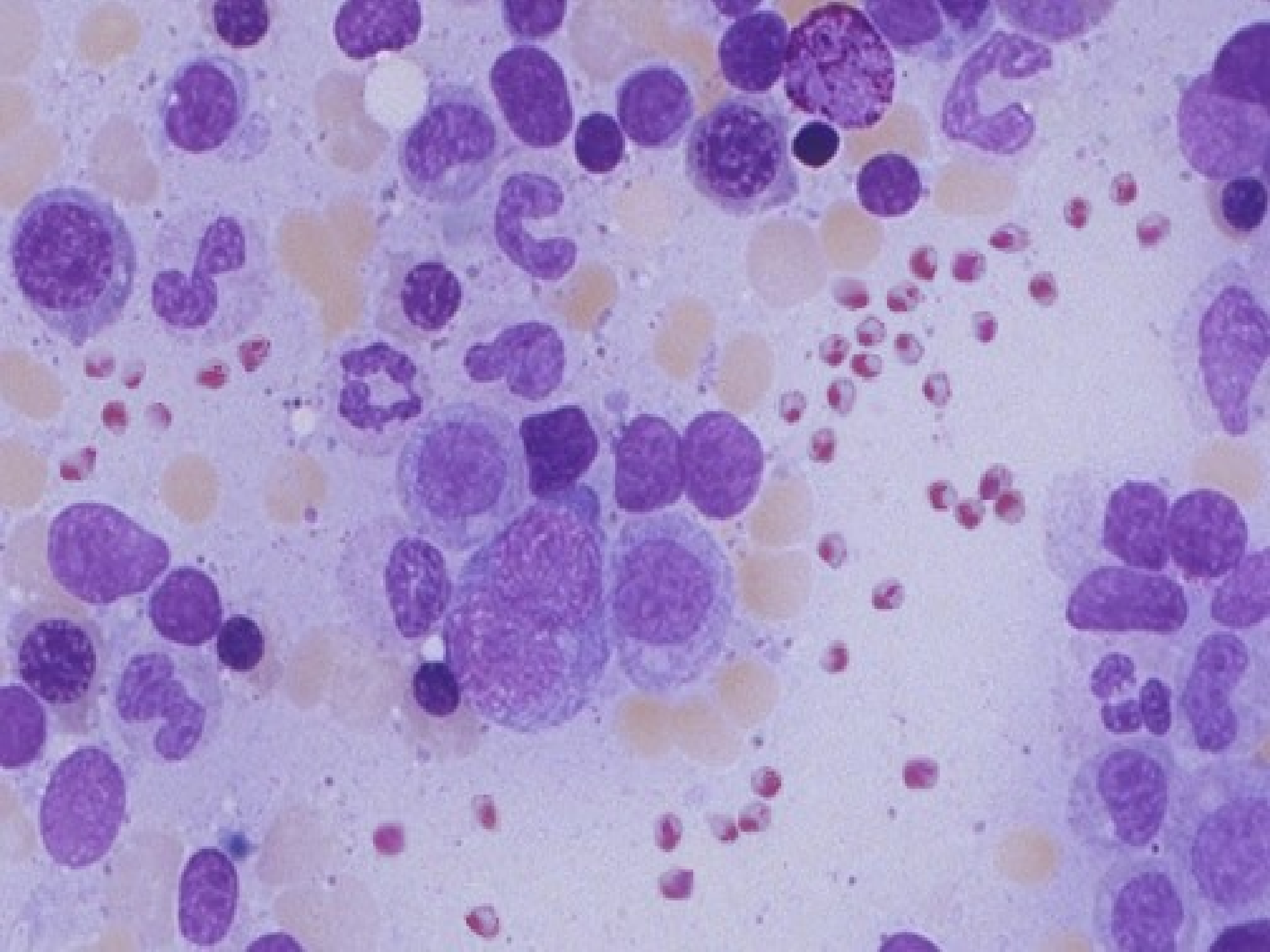


3- Another sandfly bites human and ingests blood infected with *Leishmania*

Leishmania parasite

- Stages
 - . Promastigote in sandfly
 - . Amastigote in humans





Species

- *Disease manifestation vary greatly by
parasite species*

Species

- *L. donovani*, *L. chagasi*, *L. infantum* → Visceral leishmaniasis
- *L. brasiliensis* → Mucocutaneous leishmaniasis
- *L. major*, *L. mexicana* → Cutaneous leishmaniasis
- *L. aethiopica* → Cutaneous and disseminated cutaneous leishmaniasis

pathogenesis

- Life cycle Involves two forms
 1. **Promastigote**: develops and lives extra cellularly in the vector
 2. **Amastigote**: multiplies in host macrophages
- Mammals(rodents, dogs, foxes): **reservoirs**
- **Macrophages containing amastigotes** ingested by vector=>differentiate to promastigote=> proliferate in GI =>

- Taken to salivary gland of vector=> with bite promastigotes released to dermis of host=>
- Parasite phagocytized by macrophages Acidity → round amastigotes
- Survive and reproduce in phagolysosome at PH 4.5 (the only parasite)
- Spread of amastigote depends on **host immune status** and **leishmania species**

Pathogenesis

- Spread in human body: **Depends on the leishmania species**
- Tropism appears to be temperature sensitive
 - => Parasite causing **visceral disease** grow at **37C**
 - => Parasite causing **mucocutaneous** disease grow at **34C**

Leish. Virulence Factors

- **ATPase** maintains intracellular parasite PH at 6.5
- Parasite virulence factors →
 - **Resist lyses by complement** (promastigotes in dermis)
 - **Scavenge oxygen radicals**
 - **Inhibit lysosomal enzymes** (amastigotes in macrophage)

- TH1 produces IFN gamma which activates macrophages to kill the parasite by reactive oxygen species.

Host Factors

- Severity of disease depends on host immune reaction
- Specific CMI → control infection or granulomatous inflammation
- Anergic host → No CMI → diffuse lesion

Morphology

- Four different diseases in humans
 - 1) Visceral leishmaniasis
 - 2) Cutaneous leishmaniasis
 - 3) Mucocutaneous leishmaniasis
 - 4) diffuse cutaneous leishmaniasis

Visceral Leishmaniasis

- Synonym: **Kala-azar** (Hindu word "black fever")
- Species: *L. donovani*, *L. chagasi*, *L. infantum*
- Infect macrophages in **the whole mononuclear phagocyte system**

Clinical

- Insidiously developing systemic disease
- Wide spread invasion of macrophages

Clinical presentation

- Hepato-splenomegaly
- Lymphadenopathy
- Pancytopenia

Clinical presentation

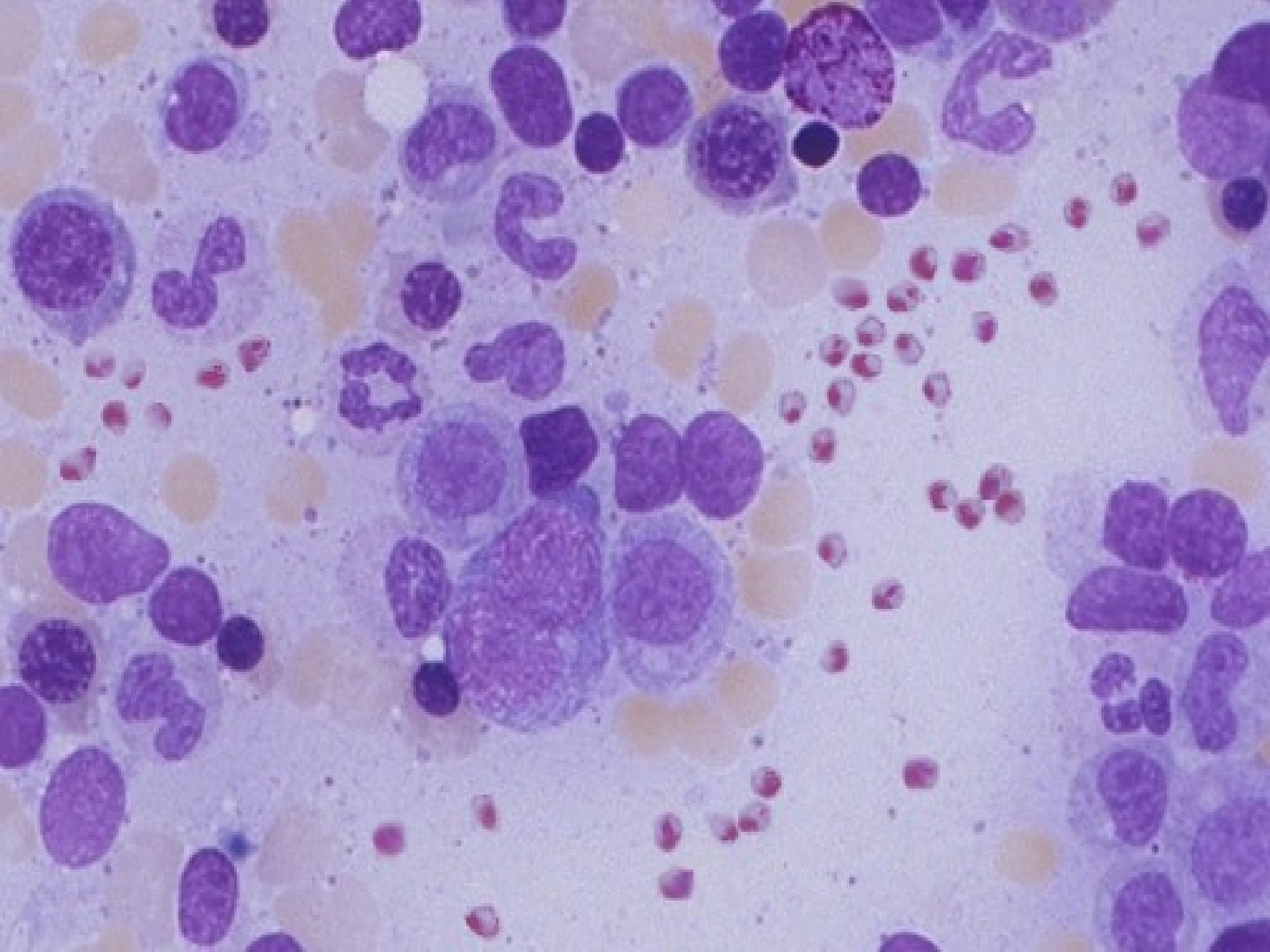
- Fever and weight loss
- Inter-current bacterial infection → usual cause of death
- Hemorrhage because of thrombocytopenia → maybe fatal

Morphology

- Splenomegaly
 - Gigantic size (3 kg)
 - Flooded with infected macrophages and plasmacytosis
- . Hepatomegally: at later stages becomes more fibrotic



- **Lymphadenopathy** → Flooded with infected macrophages and plasmacytosis
- **Bone Marrow** → crowded by infected macrophages



- Other Organs → lung, GI, pancreas etc.
- *Skin* → hyper pigmentation of extremities hence "Kala-azar"
- *Kidney* → immune complex deposition ⇒ mesangio proliferative GN
 - amyloidosis

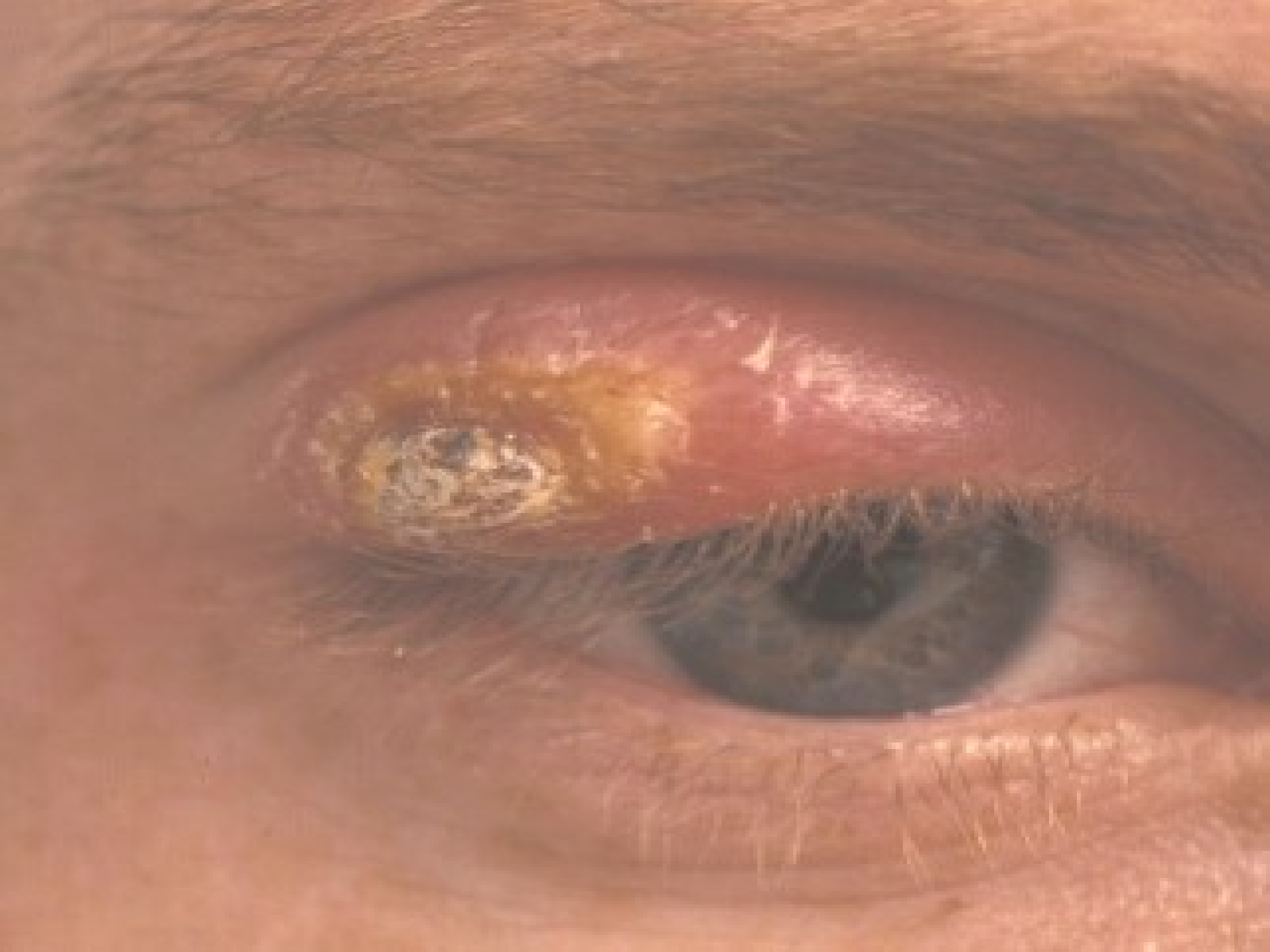
Cutaneous Leishmaniasis

- Cause: *L. major*, *L. aethiopica*
- Mild and localized disease
- Ulcer on exposed area
- Heals 6 - 18 months without treatment



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Microscopic morphology

- Granulomatous reaction
- Few parasites

Diagnosis

- FNA
- Skin biopsy
- Culture
- Skin test positive

Mucocutaneous Leishmaniasis

- Endemic in South American countries
- Moist , ulcerating, disfiguring lesions develop in nasopharynx
- Spontaneous exacerbation or remission
- Last for long time



Morphology

- Initial → parasite laden macrophages and plasma cells
- Later → granulomatous inflammation, parasite load decrease
- Eventually → Lesion remits and scars
- Reactivation → when host immunity is decreased

Disseminated Cutaneous Leishmaniasis

- Rarest form
- Found in Ethiopia, Venezuela, Brazil and Mexico

Morphology

- Sand fly bite → single skin nodule
- Nodules continue spreading
- Nodules cover entire body
- * DDX: Lepromatous leprosy







Microscopy

- Many foamy macrophages laden with parasites